1. The domain of the function $\sqrt{\log_{0.4}\frac{(x-1)}{(x+5)}\cdot\frac{1}{x^2-6}}$ **NIMCET 2007**
(a) $\{x : x < 0, x \neq -6\}$ (b) $\{x : x > 0, x \neq 6\}$
(c) $\{x : x > 1, x \neq 6\}$ (d) None of these

2. In a class, 50 student play cricket, 20 play hockey, 10 play both the games, then the no. of students playing at least one game is: **NIMCET 2007**
(a)30 (b)60 (c)70 (d)None

3. If A and B are two sets, then $A' - B'$ **NIMCET 2007**
(a)$B - A$ (b)$A - B$ (c)ϕ (d)None

4. Of the following sets, the one that includes all values of x which will satisfy $2x - 3 > 7 - x$ is: **NIMCET 2007**
(a)$x > 4$ (b)$x < \frac{10}{3}$ (c)$x = \frac{10}{3}$ (d)$x > \frac{10}{3}$

5. The range of the function $f(x) = {}^{7-x}P_{x-3}$ is **NIMCET 2007**
(a) { 1,2,3,4 } (b) { 1,2,3,4,5,6 }
(c) { 1,2,3 } (d) { 1,2,3,4,5 }

6. If $f(x) + f(1-x) = 2$, then the value of $f\left(\frac{1}{2001}\right) + f\left(\frac{2}{2001}\right) + \cdots + f\left(\frac{2000}{2001}\right)$ is **NIMCET 2008**
(a)2000 (b)2001 (c)1999 (d)1998

7. If $f(x)$ is a polynomial satisfying $f(x)\cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ and $f(3) = 28$, then $f(4)$ is given by **NIMCET 2008**
(a)63 (b)65 (c)67 (d)68

8. The number of functions f from the set $A = \{0, 1, 2\}$ into the set $B = \{0, 1, 2, 3, 4, 5, 6, 7\}$ such that $f(i) \leq f(j)$ for $i < j$ and $i, j \in A$ is **NIMCET 2008**
(a) ${}^8C_3$ (b) ${}^8C_3 + 2({}^8C_2)$
(c) ${}^{10}C_3$ (d)None

9. The total number of relations that exist from the set A with m elements into the set $A \times A$ is **NIMCET 2009**
(a)m^2 (b)m^3 (c)m (d)None

10. Set A has 3 elements and set B has 4 elements. The number of injection that can be defined from A to B is **NIMCET 2010**
(a)144 (b)12 (c)24 (d)64

11. If the function $f : [1, \infty) \to [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$, then $f^{-1}(x)$ is **NIMCET 2011**
(a)$\left(\frac{1}{2}\right)^{x(x-1)}$ (b)$\frac{1}{2}\{1 + \sqrt{1 + 4\log_2 x}\}$
(c)$\frac{1}{2}\{1 - \sqrt{1 + 4\log_2 x}\}$ (d)Not defined

12. The number of one-to-one function from {1, 2, 3} to (1, 2, 3, 4, 5} is **NIMCET 2015**
(a)125 (b)243 (c)10 (d)60

13. If the graph of $y = (x-2)^2 - 3$ is shifted by 5 units up along Y-axis and 2 units to the right along the X-axis, then the equation of the resultant graph is **NIMCET 2017**
(a)$y = x^2 + 2$ (b)$y = (x-2)^2 + 5$
(c)$y = (x+2)^2 + 2$ (d)$y = (x-4)^2 + 2$

14. Which of the following is inverse of itself? **NIMCET 2018**
(a)$f(x) = \frac{1-x}{1+x}$ (b)$f(x) = 3^{\log x}$
(c)$f(x) = 3^{x(x+1)}$ (d)None of these

15. The function $f(x) = \log\left(x + \sqrt{x^2+1}\right)$ is **NIMCET 2018**
(a) an even function (b) an odd function
(c) a periodic function
(d) Neither an even nor an odd function

16. Number of onto (surjective) functions from A to B if $n(A) = 6$ and $n(B) = 3$ is **NIMCET 2019**
(a)$2^6 - 2$ (b)$3^6 - 3$ (c)340 (d)540

17. Inverse of the function $f(x) = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$ is **NIMCET 2022**
(a)$\log_{10}(2-x)$ (b)$\frac{1}{2}\log_{10}\left(\frac{1+x}{1-x}\right)$
(c)$\frac{1}{2}\log_{10}(2x-1)$ (d)$\frac{1}{4}\log_{10}\left(\frac{2x}{2-x}\right)$

18. The function $f(x) = \log\left(x + \sqrt{x^2+1}\right)$is **NIMCET 2022**
(a)an even function (b)an odd function
(c)a periodic function
(d)Neither an even nor an odd function

19. The domain of the function $f(x) = \frac{\cos^{-1}x}{[x]}$ is **NIMCET 2022**
(a)$[-1.0) \cup \{1\}$ (b)$[-1, 1]$
(c)$[-1, 1)$ (d)None of these

20. The graph of function $f(x) = \log_e\left(x^3 + \sqrt{x^6+1}\right)$ is symmetric about **NIMCET 2023**
(a)x-axis (b)y-axis (c)origin (d)y = x

21. Let R be reflexive relation on the finite set A having 10 elements and if m is the number of ordered pair in R, then **NIMCET 2023**
(a)$m \geq 10$ (b)$m = 100$ (c)$m = 10$ (d)$m \leq 10$

22. A real valued function f is defined as $f(x) = \begin{cases} -1 & -2 \leq x \leq 0 \\ x-1 & 0 \leq x \leq 2 \end{cases}$. Which of the following statement is FALSE? **NIMCET 2023**
(a)$f(|x|) = |x| - 1$, if $0 \leq x \leq 1$
(b)$|f(x)| = x - 1$, if $1 \leq x \leq 2$
(c)$f(|x|) + |f(x)| = 1$, if $0 \leq x \leq 1$
(d)$f(|x|) = |f(x)| = 0$, if $1 \leq x \leq 2$

23. The domain of the function $\sqrt{\log_{0.4}\frac{(x-1)}{(x+5)}\cdot\frac{1}{x^2-6}}$ (Function)
(a) $\{x : x < 0, x \neq -6\}$ (b) $\{x : x > 0, x \neq 6\}$
(c) $\{x : x > 1, x \neq 6\}$ (d)none
NIMCET 2024

24. The range of the function $f(x) = {}^{7-x}P_{x-3}$ is (Function)
(a) {1,2,3,4} (b) {1,2,3,4,5,6}
(c){1,2,3} (d) {1,2,3,4,5}

NIMCET 2024

25. The value of f(1) for $f\left(\frac{1-x}{1+x}\right) = x + 2$ is (Function)
(a) 1 (b) 2
(c) 3 (d) 4

NIMCET 2024

26. If f(x) = cos [π^2] x + cos [$-\pi^2$] x, where [.] stands for the greatest integer function, then f(π/2) = (Function)
(a) –1 (b) 0
(c) 1 (d) 2

NIMCET 2024

27. The number of one-one function (Function Mapping)
f : {1,2,3} $\to$ {a, b, c, d, e} is
(a) 125 (b) 60
(c) 243 (d) None of the above

NIMCET 2025

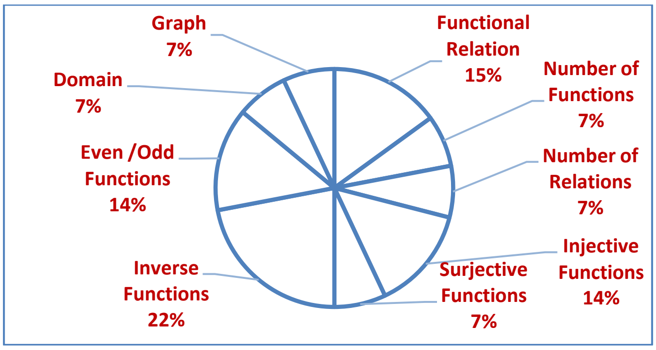
ANSWER KEYS

1.	**2.**	**3.**	**4.**	**5.**	**6.**	**7.**	**8.**	**9.**	**10.**
d	b	a	d	c	a	b	c	d	c
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b	d	d	a	b	d	b	b	a	c
21.	**22.**	**23.**	**24.**	**25.**	**26.**	**27.**			
a	c	d	c	b	a	b			

Analysis 2007 to 2025
Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	05	NIMCET 2016	00
NIMCET 2008	03	NIMCET 2017	01
NIMCET 2009	01	NIMCET 2018	02
NIMCET 2010	01	NIMCET 2019	01
NIMCET 2011	01	NIMCET 2020	00
NIMCET 2012	00	NIMCET 2021	00
NIMCET 2013	00	NIMCET 2022	03
NIMCET 2014	00	NIMCET 2023	03
NIMCET 2015	01	NIMCET 2024	04
		NIMCET 2025	01

Sub Topic Analysis



Sub Topicwise Questions Asked in NIMCET

Sub Topic	Question Asked
Function Relation	03
Number of Functions	01
Number of Relations	01
Injective Functions	02
Surjective Functions	01
Inverse Functions	03
Even/Odd Functions	03
Domain	01
Graph	02

SOLUTION

1. **(d)** Domain of $\sqrt{log_{0.4}\frac{(x-1)}{(x+5)}} \times \frac{1}{x^2-36}$

$x^2-6 \neq 0$, $x \neq 0$

$log_{0.4}\left(\frac{x-1}{x+5}\right) \geq 0$

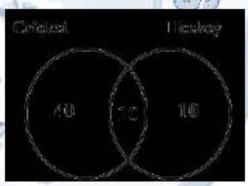
$\frac{x-1}{x+5} \leq (0.4)^0$

$\frac{x-1}{x+5} - 1 \leq 0$

$\frac{-6}{x+5} \leq 0$ $x > -5$

X : $x > -5$, $x \neq \sqrt{6}$

2. **(b)**



Student who play atleast one game = 40 + 10 + 10

3. **(a)**

$A' - B' = ?$

$A' = U - A$

$B' = U - B$

$A' - B' = (U - A) - (U - B)$

$= B - A$ option (a)

4. **(d)** x will satisfy 2x – 3 > 7 – x

2x + x > 7 + 3

3x > 10

x > 10/3

5. (c)

${}^{7-x}P_{x-3}$

x-3 $\leq$ 7 - x

2x $\leq$ 10

x $\leq$ 5 ....(i)

7x = 0 x- 3 $\geq$ 0

x $\leq$ 7 x $\geq$ 3

-(i) -(ii)

From eq (i), (ii) & (iii)

x = 3,4,5

F(3) = ${}^4P_0$, F(4) = ${}^3P_1$

F(5) = ${}^2P_2$

{1,2,3} is required range.

6. **(a)** $f(x)+f(1-x)=2$

$f\left(\frac{1}{2001}\right)+f\left(\frac{2}{2001}\right)+\cdots+f\left(\frac{2000}{2001}\right)$

$f(x)+f(1-x)=2$

$f\left(\frac{1}{2001}\right)+f\left(1-\frac{1}{2001}\right)=2 \Rightarrow f\left(\frac{1}{2001}\right)+f\left(\frac{2000}{2001}\right)=2$

$f\left(\frac{2}{2001}\right)+f\left(1-\frac{2}{2001}\right)=2 \Rightarrow f\left(\frac{2}{2001}\right)+f\left(\frac{1999}{2001}\right)=2$

$f\left(\frac{1000}{2001}\right)+f\left(1-\frac{1000}{2001}\right)=2 \Rightarrow f\left(\frac{1000}{2001}\right)+f\left(\frac{1001}{2001}\right)=2$

Total these types of pairs are 1000.

Hence, answer is $1000 \times 2 = 2000$

7. **(b)** $f(x)$ is a polynomial

such that

$f(x)\cdot f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$

In this case we should know that

$f(x)=\pm(x)^n+1$

$\Rightarrow f(3)\ =3^n+1$

$\Rightarrow 28=3^n+1 \Rightarrow 28=3^3+1$

$\Rightarrow$ n = 3

$f(4)=(4)^3+1=65$

8. **(c)** No. of ways = No. of 1-1 mappings + No. of mappings in which 0 and 1 maps to same element and 2 maps different element +0 maps to different element and 1 and 2 maps to same element + no. of constant functions

$= {}^8C_3+{}^8C_2+{}^8C_2+{}^8C_1={}^{10}C_3$

9. **(d)** Number of elements in $A\times A=m\times m$

Number of relation into set $A\times A=2^{m\times m}$

10. **(c)** $n(A)=3$ elements

$n(B)=4$ elements

Number of injections

${}^{n(B)}P_{n(A)}={}^4P_3$

$=\frac{4!}{1!}=24$

11. **(b)** $f:[1,\infty)\to[1,\infty)$

$f(x)=2^{x(x-1)}$

Let $f(x)=y$

y= $2^{x(x-1)}$

$x^2-x=\log_2 y$

$x^2-x-\log_2 y=0$

X = $\frac{1\pm\sqrt{1+4\log_2 y}}{2}$

As $x>1$, hence we select x as

$x=\frac{1}{2}\left[1+\sqrt{1+4\log_2 y}\right], x\in[1,\infty]$

Now, interchange x and y

$y=\frac{1}{2}\left[1+\sqrt{1+4\log_2 x}\right]=f^{-1}(x)$

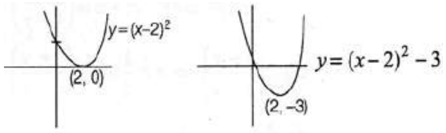
12. **(d)** One-to-one function

$n(A)=\{1,2,3\}$

$n(B)=\{1,2,3,4,5\}$

${}^5P_3==\frac{5!}{2!}=5\times4\times3=60$

13. **(d)**



When we shift this graph by 5 unit up along the Y-axis it will be $y=(x-2)^2+2$

Now, if it will shift right by two units along the X - axis new graph will be $y=(x-4)^2+2$

14. **(a)** $y=\frac{1-x}{1+x}\Rightarrow y+yx=1-x$

$\Rightarrow\ yx+x=1-y$

$\Rightarrow\ x(y+1)=1-y$

$\Rightarrow\ x=\frac{1-y}{1+y}$

Now, interchange x and y

$f^{-1}(x)=y=\frac{1-x}{1+x}$

15. **(b)** f(x) = $\log\left(x+\sqrt{x^2+1}\right)$

$f(-x)=\log\left[(-x)+\sqrt{(-x)^2+1}\right]$

$f(-x)=\log\left[\sqrt{x^2+1}-x\right]$

$=\log\left[\frac{(\sqrt{x^2+1}-x)(\sqrt{x^2+1}+x)}{\sqrt{x^2+1}+x}\right]$

$=\log\left(\frac{1}{x+\sqrt{x^2+1}}\right)=\log\left(x+\sqrt{x^2+1}\right)^{-1}$

$=-\log\left(x+\sqrt{x^2+1}\right)=-f(x)$ odd function

16. **(d)** Number of onto (surjective) function from A to B

$n(A)=6=m, n(B)=3=n$

The number of onto functions from A to B is-

$\sum_{r=1}^{n}\ (-1)^{n-r}\,{}^nC_r r^m$

$=(-1)^{3-1}\,{}^3C_1(1)^6+(-1)^{3-2}\,{}^3C_2(2)^6$

$+(-1)^{3-3}\,{}^3C_3(3)^6$

$=3-3\times64+3^6=729+3-192=732-192=540$

17. **(b)** $y=\frac{10^x-10^{-x}}{10^x+10^{-x}}=\frac{10^{2x}-1}{10^{2x}+1}$

$y\times10^{2x}+y=10^{2x}-1$

$y+1=10^{2x}(1-y)$

$10^{2x}=\frac{1+y}{1-y}\Rightarrow\log_{10}10^{2x}=\log_{10}\left(\frac{1+y}{1-y}\right)$

$\Rightarrow 2x=\log_{10}\left(\frac{1+y}{1-y}\right)$

$\Rightarrow x=\frac{1}{2}\log_{10}\left(\frac{1+y}{1-y}\right)$

$x\leftrightarrow y$

$\Rightarrow y=\frac{1}{2}\log_{10}\left(\frac{1+x}{1-x}\right)=f^{-1}(x)$

18. **(b)** Refer to solution 10.

19. **(a)** $\cos^{-1} x$

Here, $-1 \le x \le 1$

Now, $[x] \ne 0$

Hence, x can not lie between $0 \le x < 1$.

Domain of function is $-1 \le x < 0 \cup x = 1$

$x \in [-1,0) \cup \{1\}$

20. **(c)** $f(x) = \log\left(x^3 + \sqrt{x^6+1}\right)$

$f(-x) = \log\left[(-x)^3 + \sqrt{(-x)^6+1}\right]$

$f(-x) = \log\left[\sqrt{x^6+1} - x^3\right]$

$= \log\left[\frac{(\sqrt{x^6+1}-x^3)(\sqrt{x^6+1}+x^3)}{\sqrt{x^6+1}+x^3}\right]$

] $= \log\left(\frac{1}{x^3+\sqrt{x^6+1}}\right) = \log\left(x^3+\sqrt{x^6+1}\right)^{-1}$

$= -\log\left(x^3+\sqrt{x^6+1}\right) = -f(x)$ Odd function

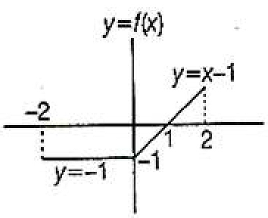
We should know that odd functions are symmetrical about origin.

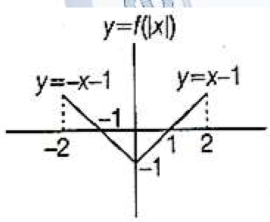
21. **(a)** Given R has m order pairs.

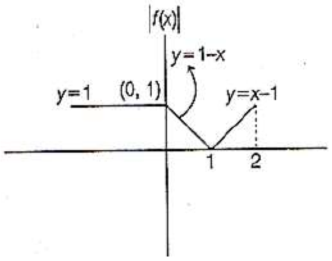
Since, R is reflexive relation on A, therefore $(a,a) \in R \forall a \in A$

Then the minimum no. of ordered pairs in R is 10 . Therefore, $m \ge 10$

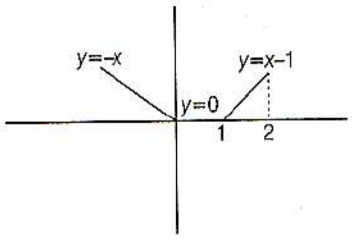
22. **(c)** Graphical solution







Add these two graphs y = |f(x) + f(|x|)



From given graphs we can find that statement (c) is false.

23. **(d)** Domain of $\sqrt{log_{0.4}\frac{(x-1)}{(x+5)}} \times \frac{1}{x^2-36}$

$x^2–6 \ne 0$, $x \ne 0$

$log_{0.4}\left(\frac{x-1}{x+5}\right) \ge 0$

$\frac{x-1}{x+5} \le (0.4)^0$

$\frac{x-1}{x+5} - 1 \le 0$

$\frac{-6}{x+5} \le 0$ $x > –5$

X : x > –5, $x \ne \sqrt{6}$

24. **(c)**

$^{7-x}P_{x-3}$

$x-3 \le 7 - x$

$2x \le 10$

$x \le 5$ ....(i)

$7x = 0$ $x- 3 \ge 0$

$x \le 7$ $x \ge 3$

-(i) -(ii)

From eq (i), (ii) & (iii)

x = 3,4,5

$F(3) = {}^4P_0$, $F(4) = {}^3P_1$

$F(5) = {}^2P_2$

{1,2,3} is required range.

25. (b) The value of f(1) for $f\left(\frac{1-x}{1+x}\right) = x + 2$ is

let x = 0

$f\left(\frac{1-0}{1+0}\right) = 0 + 2$

f(1) = 2

26. (a) f(x) = cos $[\pi^2]$ x + cos $[-\pi^2]$ x [.] →GIF

f(x) = cos [9.85] x + cos [-9.85]x

f(x) = cos 9x + cos 10 x

$f\left(\frac{\pi}{2}\right) = cos\frac{9}{2}\pi + cos5\pi$

= 0 – 1

= –1

27. (b) f : {1,2,3} → {a, b, c, d,e}

Total one-one

= 5 x 4 x 3 = 60